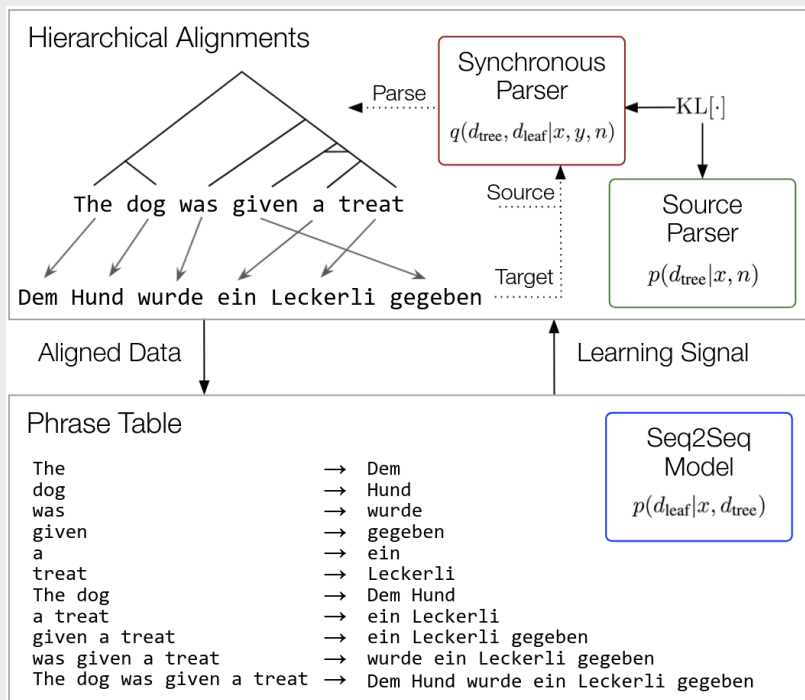


Hierarchical Phrase-based Sequence-to-Sequence Learning

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Model Overview



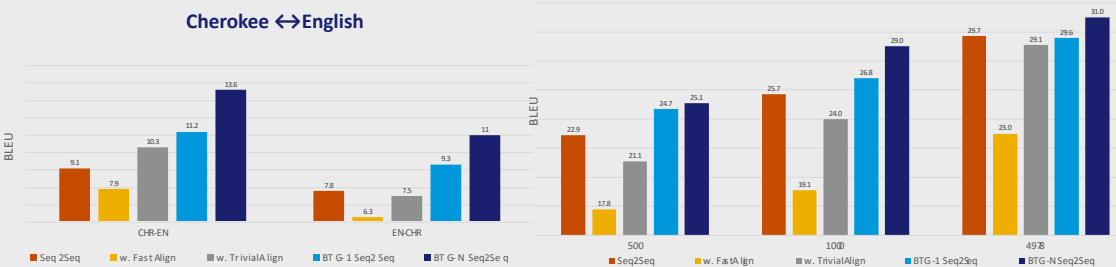
- During training, a *variational synchronous parser* hierarchically segments and aligns source-target phrases (top), which provides training pairs for a seq2seq model (bottom).
- The seq2seq loss provides *learning signal to the synchronous parser* which subsequently provides signal to a monolingual source parser.
- After training, there are **two inference models**: one mode in which both parsers are discarded, which results in a *regular seq2seq model* that has been regularized by latent hierarchical phrase pairs; another mode in which the seq2seq model can be combined with the source parser to *derive a neural synchronous grammar*.

Generative Process $p(d|x) = p(n|x)p(d_{tree}|x, n)p(d_{leaf}|x, d_{tree})$

$p(n x)$ # Segments	Source Parser: $p(d_{tree} x, n)$ Nonterminal Derivations (Source Segmentation & Reordering)	Seq2Seq: $p(d_{leaf} x, d_{tree})$ Leaf Derivations (Phrase-to-Phrase Translations)
1	The dog was given a treat	Dem Hund wurde ein Leckerli gegeben
2	The dog was given a treat	Dem Hund wurde ein Leckerli gegeben
3	The dog was given a treat	Dem Hund wurde ein Leckerli gegeben
4	The dog was a treat given	Dem Hund wurde ein Leckerli gegeben
⋮	⋮	⋮
6	The dog was a treat given	Dem Hund wurde ein Leckerli gegeben

- Sample the number of segments in the target $n \sim p(n|x)$
- Sample the tree topology which gives a segmentation and reordering $d_{tree} \sim p(d_{tree}|x, n)$
- Sample leaf derivations (phrase-to-phrase translations) from a seq2seq model $d_{leaf} \sim p(d_{leaf}|x, d_{tree})$

Results on Machine Translation



Constrained Decoding

Input:	Die europäische Krise <i>schließt den Kreis</i>
New translation rule:	<i>schließt den Kreis</i> → is coming full circle
Original Prediction:	The euro crisis closes the loop
New Prediction:	The European crisis is coming full circle
Reference:	The European crisis is coming full circle
Google Translate:	The European crisis closes the circle
Input:	Die ARD strahlte gestern um 20:15 "Aus dem Nichts" aus
New translation rule:	"Aus dem Nichts" → "In the Fade", 20:15 → 8.15 pm
Original Prediction:	The ARD aired yesterday at 20:15 "Out of Nowhere"
New Prediction:	The ARD aired yesterday 8.15 pm "In the Fade"
Reference:	ARD broadcast "In the Fade" yesterday at 8.15 pm
Google Translate:	Yesterday at 8:15 p.m., ARD broadcast "Out of Nowhere".

Induced Alignments

